Metric
DOD-S-63275/1A
29 October 1982
SUPERSEDING
DOD-S-63275/1
26 September 1977

MILITARY SPECIFICATION SHEET

STUD, SCREW THREAD, LOCKED IN, RING LOCKED, SERRATED, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the stud described herein shall consist of this specification and the latest issue of specification DOD-S-63275.

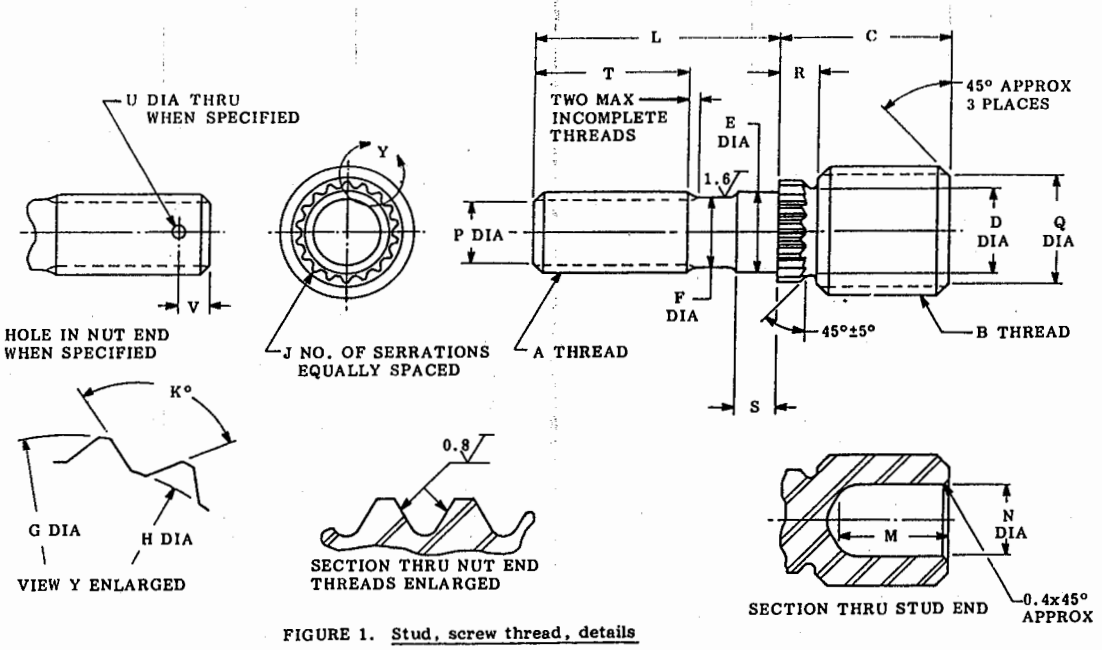


FIGURE 1. Stud, screw thread, details

NOTES:

1. Dimensions are in millimeters, to be met after plating.
2. Concentricity: Shank of nut end shall be concentric with serrated collar within 0.15 FIM.
3. Fillets shall be 0.4 R maximum.
4. Edges shall be broken 0.1 - 0.4 unless otherwise specified.

TABLE I. Part dimensions, short stud end length

DASH NO.	A NUT END THREAD SIZE (SEE REQT 5)	B STUD END THREAD	C ±0.45	D DIA MIN	E DIA +0.0 -0.2	F DIA APPROX	G DIA +0.0 -0.2	H DIA MAX	J	K° +2° -1°	M REF	N DIA APPROX	P DIA ±0.2	Q DIA ±0.2	R ±0.3	S ±0.5	T MIN	U DIA +0.1 -0.0	V +0.0 -0.3	BASIC LOCKRING
101	5x0.8	M8x1.25	9.3	4.72	5.0	4.5	6.02	5.26	13	102°	5.5	4.0	3.68	6.07	2.8	2.5	12.0	1.0	1.9	DOD-I-63276/2-1
102	6x1	M10x1.5	10.8	5.65	6.0	5.3	7.40	6.57	17	86°	6.0	5.5	4.40	7.73	3.4	3.0	14.0	1.5	2.8	DOD-I-63276/2-2
103	7x1	M12x1.5	12.3	7.57	7.0	6.3	9.45	8.51	17	99°	6.5	7.5	5.40	9.73	4.0	3.5	15.0	1.5	2.8	DOD-I-63276/2-3
104	8x1.25	M12x1.5	14.5	7.57	8.0	7.2	9.45	8.51	17	99°	9.0	6.5	6.07	9.73	4.0	4.0	18.0	2.0	3.5	DOD-I-63276/2-3
105	8x1	M12x1.5	14.5	7.57	8.0	7.3	9.45	8.51	17	99°	9.0	6.5	6.40	9.73	4.0	4.0	18.0	2.0	3.5	DOD-I-63276/2-3
106	10x1.5	M14x1.5	18.0	9.48	10.0	9.0	11.61	10.59	21	97°	12.0	6.7	7.73	11.73	4.6	5.0	20.0	2.0	3.5	DOD-I-63276/2-4
107	10x1.25	M14x1.5	18.0	9.48	10.0	9.2	11.61	10.59	21	97°	12.0	6.7	8.07	11.73	4.6	5.0	20.0	2.0	3.5	DOD-I-63276/2-4
108	12x1.5	M16x1.5	22.5	11.48	12.0	11.0	13.21	12.22	24	96°	15.0	7.5	9.73	13.73	5.6	6.0	22.0	3.0	4.5	DOD-I-63276/2-5
109	12x1.25	M16x1.5	22.5	11.48	12.0	11.2	13.21	12.22	24	96°	15.0	7.5	10.06	13.73	5.6	6.0	22.0	3.0	4.5	DOD-I-63276/2-5
110	14x1.5	M20x1.5	24.0	13.48	14.0	13.0	15.45	14.48	26	111°	15.5	11.5	11.73	17.73	6.0	7.0	24.0	3.0	5.0	DOD-I-63276/2-6

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TABLE II. Part dimensions, long stud end length

DASH NO.	A NUT END THREAD SIZE (SEE REQT 5)	B STUD END THREAD	C ±0.45	D DIA MIN	E DIA +0.0 -0.2	F DIA APPROX	G DIA +0.0 -0.2	H DIA MAX	J	K° +2° -1°	M REF	N DIA APPROX	P DIA ±0.2	Q DIA ±0.2	R ±0.3	S ±0.5	T MIN	U DIA +0.1 -0.0	V +0.0 -0.3	BASIC LOCKRING
201	5x0.8	M8x1.25	12.2	4.72	5.0	4.5	6.02	5.26	13	102°	8.5	4.0	3.68	6.07	2.8	2.5	12.0	1.0	1.9	DOD-I-63276/2-1
202	6x1	M10x1.5	14.1	5.65	6.0	5.3	7.40	6.57	17	86°	9.5	5.5	4.40	7.73	3.4	3.0	14.0	1.5	2.8	DOD-I-63276/2-2
203	7x1	M12x1.5	16.2	7.57	7.0	6.3	9.45	8.51	17	99°	10.5	7.5	5.40	9.73	4.0	3.5	15.0	1.5	2.8	DOD-I-63276/2-3
204	8x1.25	M12x1.5	19.8	7.57	8.0	7.2	9.45	8.51	17	99°	14.5	6.5	6.07	9.73	4.0	4.0	18.0	2.0	3.5	DOD-I-63276/2-3
205	8x1	M12x1.5	19.8	7.57	8.0	7.3	9.45	8.51	17	99°	14.5	6.5	6.40	9.73	4.0	4.0	18.0	2.0	3.5	DOD-I-63276/2-3
206	10x1.5	M14x1.5	25.1	9.48	10.0	9.0	11.61	10.59	21	97°	19.0	6.7	7.73	11.73	4.6	5.0	20.0	2.0	3.5	DOD-I-63276/2-4
207	10x1.25	M14x1.5	25.1	9.48	10.0	9.2	11.61	10.59	21	97°	19.0	6.7	8.07	11.73	4.6	5.0	20.0	2.0	3.5	DOD-I-63276/2-4
209	12x1.5	M16x1.5	31.9	11.48	12.0	11.0	13.21	12.22	24	96°	24.5	7.5	9.73	13.73	5.6	6.0	22.0	3.0	4.5	DOD-I-63276/2-5
29	12x1.25	M16x1.5	31.9	11.48	12.0	11.2	13.21	12.22	24	96°	24.5	7.5	10.06	13.73	5.6	6.0	22.0	3.0	4.5	DOD-I-63276/2-5
210	14x1.5	M20x1.5	34.2	13.48	14.0	13.0	15.45	14.48	26	111°	26.0	11.5	11.73	17.73	6.0	7.0	24.0	3.0	5.0	DOD-I-63276/2-6

TABLE III. Available nut end lengths, mass values

AVAILABLE NUT END LENGTHS 1/ L	MASS kg/1000 PIECES APPROX 2/													
	M5x0.8		M6x1		M7x1		M8x1.25 M8x1		M10x1.5 M10x1.25		M12x1.5 M12x1.25		M14x1.5	
	101	201	102	202	103	203	104 105	204 205	106 107	206 207	108 109	208 209	110	210
18	4.05	4.64	--	--	--	--	--	--	--	--	--	--	--	--
20	4.29	4.88	6.72	7.70	--	--	--	--	--	--	--	--	--	--
22	4.53	5.12	7.06	8.04	9.84	11.3	--	--	--	--	--	--	--	--
25	4.89	5.48	7.57	8.55	10.60	12.0	14.8	17.3	--	--	--	--	--	--
28	5.25	5.84	8.09	9.07	11.30	12.7	15.8	18.2	28.4	32.8	--	--	--	--
30	5.49	6.08	8.43	9.41	11.80	13.2	16.4	18.8	29.4	33.8	--	--	--	--
(32)	5.73	6.32	8.77	9.75	12.30	13.7	17.0	19.4	30.4	34.8	47.4	57.1	--	--
35	6.09	6.68	9.29	10.30	13.00	14.4	17.9	20.4	31.8	36.3	49.6	59.3	71.7	85.7
(38)	6.45	7.04	9.80	10.80	13.70	15.1	18.9	21.3	33.3	37.8	51.8	61.5	74.8	88.8
40	6.69	7.28	10.20	11.10	14.20	15.6	19.5	22.0	34.3	38.8	53.3	63.0	76.9	90.9
(42)	6.93	7.52	10.50	11.50	14.70	16.1	20.1	22.6	35.3	39.7	54.8	64.5	78.9	92.9
45	7.29	7.88	11.00	12.00	15.40	16.8	21.1	23.5	36.8	41.2	57.0	66.7	82.0	96.0
(48)	7.65	8.24	11.50	12.50	16.20	17.6	22.0	24.4	38.3	42.7	59.2	68.9	85.1	99.1
50	7.89	8.48	11.90	12.80	16.60	18.1	22.6	25.1	39.2	43.7	60.7	70.4	87.2	101.0
(52)	8.13	8.72	12.20	13.20	17.10	18.5	23.2	25.7	40.2	44.7	62.2	71.9	89.3	103.0
55	8.49	9.08	12.70	13.70	17.90	19.3	24.2	26.6	41.7	46.2	64.4	74.1	92.4	106.0
(58)	8.85	9.44	13.20	14.20	18.60	20.0	25.1	27.6	43.2	47.6	66.6	76.3	95.5	109.0
60	9.09	9.68	13.60	14.60	19.10	20.5	25.8	28.2	44.2	48.6	68.1	77.8	97.6	112.0
(62)	9.33	9.92	13.90	14.90	19.60	21.0	26.4	28.8	45.2	49.6	69.6	79.3	99.6	114.0
65	9.69	10.30	14.30	15.40	20.30	21.7	27.3	29.8	46.6	51.1	71.8	81.5	103.0	117.0
(68)	10.10	10.60	14.90	15.90	21.00	22.4	28.2	30.7	48.1	52.6	74.0	83.7	106.0	120.0
70	10.30	10.90	15.30	16.30	21.50	22.9	28.9	31.3	49.1	53.5	75.5	85.2	108.0	122.0
(72)	10.50	11.10	15.60	16.60	22.00	23.4	29.5	32.0	50.1	54.5	77.0	86.7	110.0	124.0
75	10.90	11.50	16.10	17.10	22.70	24.1	30.4	32.9	51.6	56.0	79.2	88.9	113.0	127.0
(78)	11.20	11.80	16.70	17.60	23.40	24.9	31.4	33.8	53.0	57.5	81.4	91.1	116.0	130.0
80	11.50	12.10	17.00	18.00	23.90	25.4	32.0	34.4	54.0	58.5	82.9	92.6	118.0	132.0
85	--	--	17.90	18.80	25.10	26.6	33.6	36.0	56.5	60.9	86.6	96.3	123.0	137.0
90	--	--	18.70	19.70	26.40	27.8	35.1	37.6	59.0	63.4	90.3	100.0	129.0	143.0
95	--	--	--	--	27.60	29.0	36.7	39.1	61.4	65.9	94.0	104.0	134.0	148.0
100	--	--	--	--	28.80	30.2	38.2	40.7	63.9	68.3	97.7	107.0	139.0	153.0
105	--	--	--	--	--	--	39.8	42.3	66.4	70.8	101.0	111.0	144.0	158.0
110	--	--	--	--	--	--	41.4	43.8	68.8	73.3	105.0	115.0	149.0	163.0

Sizes shown in brackets are non-standard lengths

1/ Lengths over 110mm increase in increments of 5mm

2/ Calculated with 7.9 kg/mm^3

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REQUIREMENTS:

1. Material: Steel, alloy, composition 8740 (UNS G87400) per AMS 6322. Nickel base alloy, corrosion and heat resistant composition 718 (UNS N07718) per AMS 5662. (See material code letters per Requirement 4).

2. Protective coating or treatment: Steel, UNS G87400 Code A shall be cadmium plated per general specification. Steel, UNS N07718, Code B shall be cleaned and descaled per general specification.

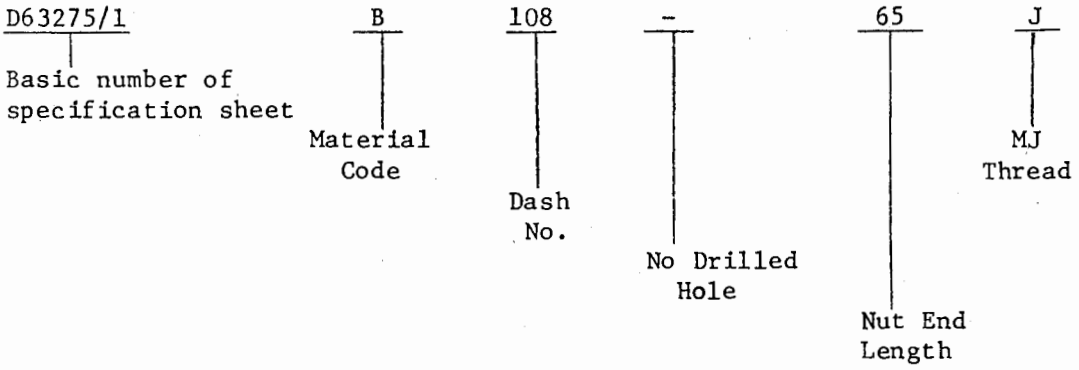
3. Surface texture: 3.2 µm except serrations.

4. Mechanical properties:

Material Code Letter	Material	Hardness, Min.	Min. Tensile Strength MPa	Dash Numbers
A	Steel, Alloy, Comp. 8740	28HRC	900	-101 thru -110 &-201 thru -210
B	Nickel Base Alloy, Comp. 718	40HRC	1250	-101 thru -110

5. Military part number: Consists of the letter D, plus the basic number of the specification sheet, plus material code (A or B) plus dash number from Table I or Table II, plus letter D for drilled hole in nut end or use dash if no drilled hole is required, plus nut end length per Table III, plus letter J for MJ thread on nut end or leave blank if M thread is specified.

Example of Military part number:



Explanation of part number example:

Stud, nickel base alloy, MJ12x1.5 thread size, no drilled hole in nut end, and 65mm nut end length.

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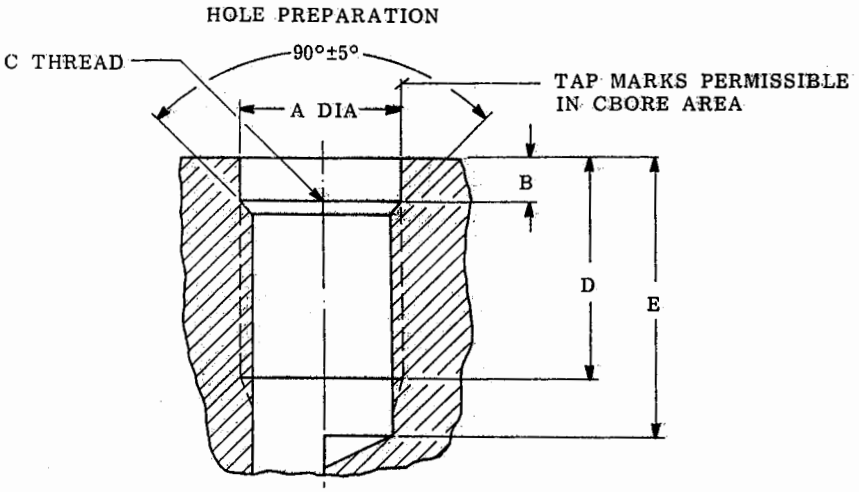


FIGURE 2. Parent material hole

TABLE IV. Parent material hole dimensions

NOMINAL NUT END THD SIZE OF STUD	STUD DASH NUMBER	A DIA	B +0.4 -0.0	C THREAD -4H5H	D MIN FULL THREAD DEPTH	E MIN RECMD DRILL DEPTH BLIND HOLE
M5x0.8	101	8.80- 8.90	2.4	M8x1.25	10.2	13.6
M5x0.8	201	8.80- 8.90	2.4	M8x1.25	13.1	16.5
M6x1	102	10.00-10.13	3.0	M10x1.5	11.7	15.3
M6x1	202	10.00-10.13	3.0	M10x1.5	15.0	18.6
M7x1	103	12.00-12.13	4.0	M12x1.5	13.2	16.8
M7x1	203	12.00-12.13	4.0	M12x1.5	17.1	20.7
M8x1.25	104	12.00-12.13	4.0	M12x1.5	15.4	19.0
M8x1.25	204	12.00-12.13	4.0	M12x1.5	20.7	24.3
M8x1	105	12.00-12.13	4.0	M12x1.5	15.4	19.0
M8x1	205	12.00-12.13	4.0	M12x1.5	20.7	24.3
M10x1.5	106	14.00-14.15	4.7	M14x1.5	18.9	22.5
M10x1.5	206	14.00-14.15	4.7	M14x1.5	26.0	29.6
M10x1.25	107	14.00-14.15	4.7	M14x1.5	18.9	22.5
M10x1.25	207	14.00-14.15	4.7	M14x1.5	26.0	29.6
M12x1.5	108	16.00-16.15	5.7	M16x1.5	23.4	27.0
M12x1.5	208	16.00-16.15	5.7	M16x1.5	32.8	36.4
M12x1.25	109	16.00-16.15	5.7	M16x1.5	23.4	27.0
M12x1.25	209	16.00-16.15	5.7	M16x1.5	32.8	36.4
M14x1.5	110	20.00-20.18	6.0	M20x1.5	24.9	28.5
M14x1.5	210	20.00-20.18	6.0	M20x1.5	35.1	38.7

NOTES:

1. Diameter "A" and thread shall be concentric within 0.1 FIM.
2. Axis of hole shall be normal to entry surface or provide spot face when required.
3. Machined surfaces shall be 3.2 µm in accordance with ANSI B46.1.
4. Dimensions are in millimeters.

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5. Stud installation procedure.

5.1 Wrench stud to depth as shown in figure 3.

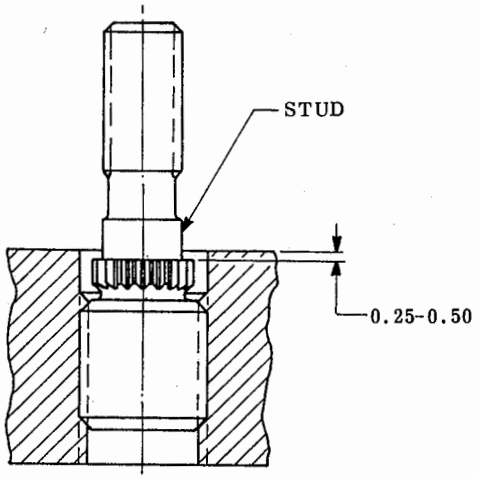


FIGURE 3. Stud wrenched-in

5.2 Drive lockring to depth shown in figure 4.

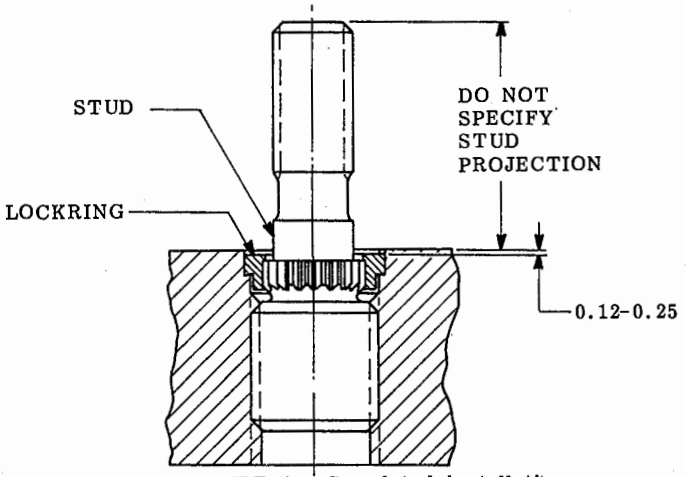


FIGURE 4. Completed installation

5.3 Replacement of studs is made with same size parts as those removed and in same manner as originally installed. Rotate new lockring if necessary, before driving, to a position in which the external serrations are aligned with those in parent material.

5.4 Use of manufacturers installation tools mandatory (Rosan, Inc. Newport Beach, CA - FSCM 83324).

Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - AR
Navy - AS

Preparing activity:
Army - AR

Project No. 5307-0321

Review activities:
Army - AT, AV, MI, ER, GR
Navy - SH
NSA - NS
DLA - IS

User activities:
Army - ME
Navy - OS, MC

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